Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/16/2016
PI Name:	Moore, Steven T. Ph.D.		
Project Title:	Assessment of Operator Proficiency Following Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10029	**Congressional District:**	14
Comments:	NOTE: PI moved to Central Queensland University, Australia, July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	09/06/2012	**End Date:**	07/03/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/3/2016 (original end date 8/15/2016) due to PI move to Central Queensland University, Australia, per HRP (Ed., 8/26/16)		
Key Personnel Changes/Previous PI:	Co-Principal Investigator is Hamish MacDougall/University of Sydney (Australia).		
COI Name (Institution):	MacDougall, Hamish (University of Sydney (Australia))		
Grant/Contract No.:	NNX12AM25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR'S NOTE (3/5/2013): In order to continue work on the flight phase of this project, this new award (NNX12AM25G) began 9/6/2012. See project with same title (grant NNX09AL14G) for previous reporting. Evidence from shuttle landings suggests that performance of astronaut pilots was degraded after microgravity exposure (Moore et al., 2008). The underlying mechanism is unclear, but a major contributing factor is likely central adaptation to the relative absence of gravity on orbit, thought to be responsible for a range of sensorimotor deficits (postural, locomotor, oculomotor, perception of motion) observed in astronauts post-landing. Long-term exposure to microgravity has the potential to negatively impact the ability of crewmembers to navigate and land the crew exploration vehicle and perform post-landing surface operations for exploration class missions (asteroid/Mars). In this proposal we addressed the 'risk of impaired ability to maintain control of vehicles and other complex systems' identified in the Human Research Program (HRP) Human Research Roadmap, in particular the sensorimotor gap SM6.1 to 'determine if sensorimotor dysfunction during and after long-duration spaceflight affects ability to control spacecraft and associated systems'. Our team sought to answer four questions implicit in SM6.1: 1. To what degree does long-duration spaceflight impair a crewmember's ability to operate a vehicle or other machinery? 2. What sensorimotor/cognitive functions underlie degradation of operator proficiency? 3. Are sensorimotor countermeasures required for asteroid/Martian landings and surface operations? 4. If so, what areas should these countermeasures target? This project utilized motion-based simulations of operationally relevant tasks (driving, flight, rover) to assess the impact of spaceflight on operator proficiency in a group of 8 astronauts, as well as a battery of cognitive/sensorimotor tests to determine the underlying cause of post-flight performance decrements. In addition, a ground control (shadow) group (N=12) and a sleep restriction cohort (N=8) were also tested to control for non-spaceflight factors such as lack of practice between pre- and post-flight testing and sleepiness. The results demonstrated specific deficits on the day of landing in manual dexterity, dual tasking and motion perception, and significant degradation in operator proficiency, that were unique to the astronaut group. Based on these findings, we provide countermeasure recommendations for future consideration by the HRP.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The techniques developed as part of this grant have potential application to assessment and rehabilitation of patients with a variety of neurological conditions, such as stroke, traumatic brain injury, and vestibular disease.
Task Progress:	Significant decrements in the ability to operate vehicles were observed in eight astronauts returning from 6-months aboard the International Space Station (ISS). Full-motion simulations were performed within 24-hours of landing. Compared to performance prior to launch, the ability to maintain lane position when driving was significantly impaired; all astronauts made more crossings into the wrong lane and were slower to correct. Piloting performance (landing a T-38 jet) in six astronauts was compromised on landing day, including one crash at the runway threshold. Variability in performance during a Mars rover docking maneuver was also significantly increased. Performance in a shadow group (tested at the same time points) and a sleep cohort (tested after 30 hours of sleep deprivation) was unchanged, demonstrating that the deficits observed in astronaut proficiency were due to spaceflight. There were subtle but significant changes in cognitive and sensorimotor performance on the day of return. There was a 10% decline in manual dexterity and perception of cabin tilt was diminished, suggesting a general post-flight malaise in motor function and perception. Although tracking a moving target with a mouse was unaffected, adding a distracting task (entering 4-digit codes with the non-tracking hand) significantly impaired performance, indicating post-flight limitations in shareable central processing resources; a lack of cognitive reserve apparent only when faced with competing tasks. These changes were not observed in the shadow or sleep-restricted groups. Performance on all tasks returned to baseline by 4-days post-landing. Taken together, the results suggest that a range of subtle physiological changes in spaceflight combine to leave astronauts particularly vulnerable to performance decrements on the day of landing, particularly when faced with low-frequency gravito-inertial or visual tilts and multiple and competing task requirements. Suggested countermeasures include 'just-in-time' training, vehicle tilt displays, self-assessments to gauge fitness prior to challenging tasks, and limitations in dual-tasking during manual control.
Bibliography Type:	Description: (Last Updated: 09/07/2020)
Articles in Peer-reviewed Journals	Yungher DA, Morris TR, Dilda V, Shine JM, Naismith SL, Lewis SJ, Moore ST. "Temporal characteristics of high-frequency lower-limb oscillation during freezing of gait in Parkinson's disease." Parkinsons Dis. 2014;2014:606427. Epub 2014 Jul 2. http://dx.doi.org/10.1155/2014/606427 ; PubMed PMID: 25101189; PubMed Central PMCID: PMC4101926 , Aug-2014
Articles in Peer-reviewed Journals	Dilda V, Morris TR, Yungher DA, MacDougall HG, Moore ST. "Central adaptation to repeated galvanic vestibular stimulation: implications for pre-flight astronaut training." PLoS One. 2014 Nov 19;9(11):e112131. http://dx.doi.org/10.1371/journal.pone.0112131 ; PubMed PMID: 25409443; PubMed Central PMCID: PMC4237321 , Nov-2014
Articles in Peer-reviewed Journals	Moore ST, Dilda V, Morris TR, Yungher DA, MacDougall HG. "Pre-adaptation to noisy Galvanic vestibular stimulation is associated with enhanced sensorimotor performance in novel vestibular environments." Front Syst Neurosci. 2015 Jun 8;9:88. http://dx.doi.org/10.3389/fnsys.2015.00088 ; PubMed PMID: 26106308; PubMed Central PMCID: PMC4458607 , Jun-2015
Articles in Peer-reviewed Journals	Hallgren E, Migeotte PF, Kornilova L, Delière Q, Fransen E, Glukhikh D, Moore ST, Clément G, Diedrich A, MacDougall H, Wuyts FL. "Dysfunctional vestibular system causes a blood pressure drop in astronauts returning from space." Sci Rep. 2015 Dec 16;5:17627. http://dx.doi.org/10.1038/srep17627 ; PubMed PMID: 26671177; PubMed Central PMCID: PMC4680856 , Dec-2015

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Articles in Peer-reviewed Journals	Moore ST, Yungher DA, Morris TR, Dilda V, MacDougall HG, Shine JM, Naismith SL, Lewis SJ. "Autonomous identification of freezing of gait in Parkinson's disease from lower-body segmental accelerometry." J Neuroeng Rehabil. 2013 Feb 13;10:19. http://dx.doi.org/10.1186/1743-0003-10-19 ; PubMed PMID: 23405951; PubMed Central PMCID: PMC3598888 , Feb-2013
Articles in Peer-reviewed Journals	Gilat M, Hall JM, Ehgoetz Martens KA, Shine JM, Walton CC, MacDougall HG, Moore ST, Lewis SJ. "Staircase climbing is not solely a visual compensation strategy to alleviate freezing of gait in Parkinson's disease." J Neurol. 2017 Jan;264(1):174-6. Epub 2016 Nov 29. http://dx.doi.org/10.1007/s00415-016-8329-3 ; PubMed PMID: 27900497 , Jan-2017
Articles in Peer-reviewed Journals	Moore ST, Dilda V, Morris TR, Yungher DA, MacDougall HG, Wood SJ. "Long-duration spaceflight adversely affects post-landing operator proficiency." Sci Rep. 2019 Feb 25;9(1):2677. https://doi.org/10.1038/s41598-019-39058-9 ; PubMed PMID: 30804413; PubMed Central PMCID: PMC6389907 , Feb-2019